

Hands on 1

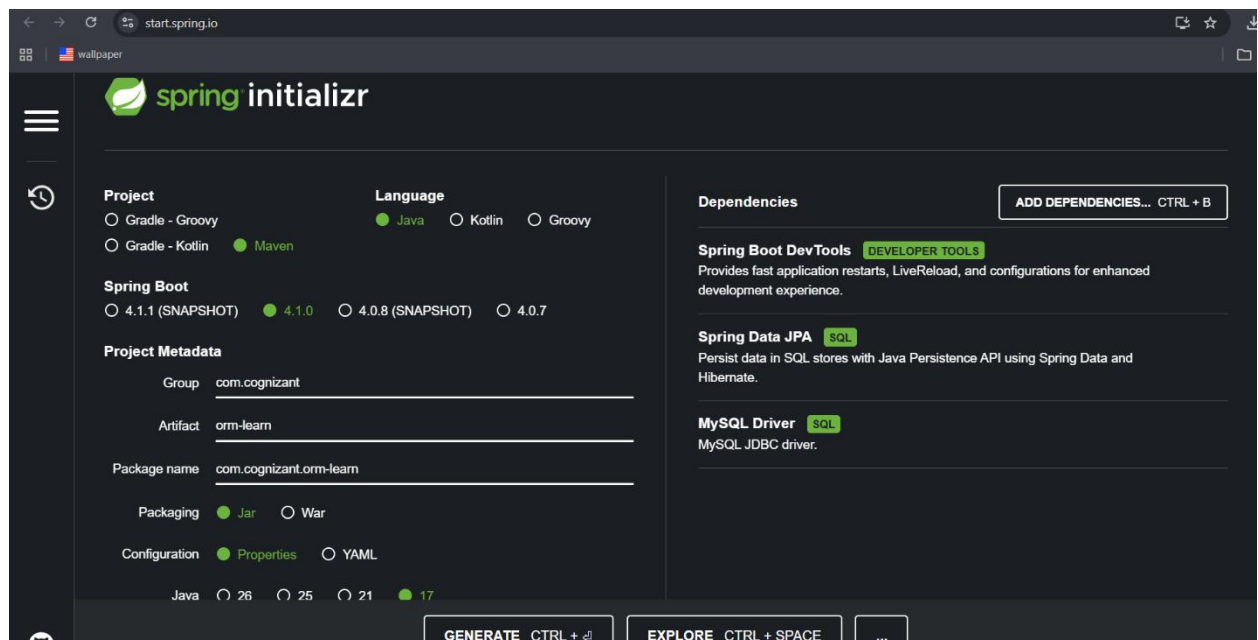
Spring Data JPA - Quick Example

Software Pre-requisites

- MySQL Server 8.0
- MySQL Workbench 8
- Eclipse IDE for Enterprise Java Developers 2019-03 R
- Maven 3.6.2

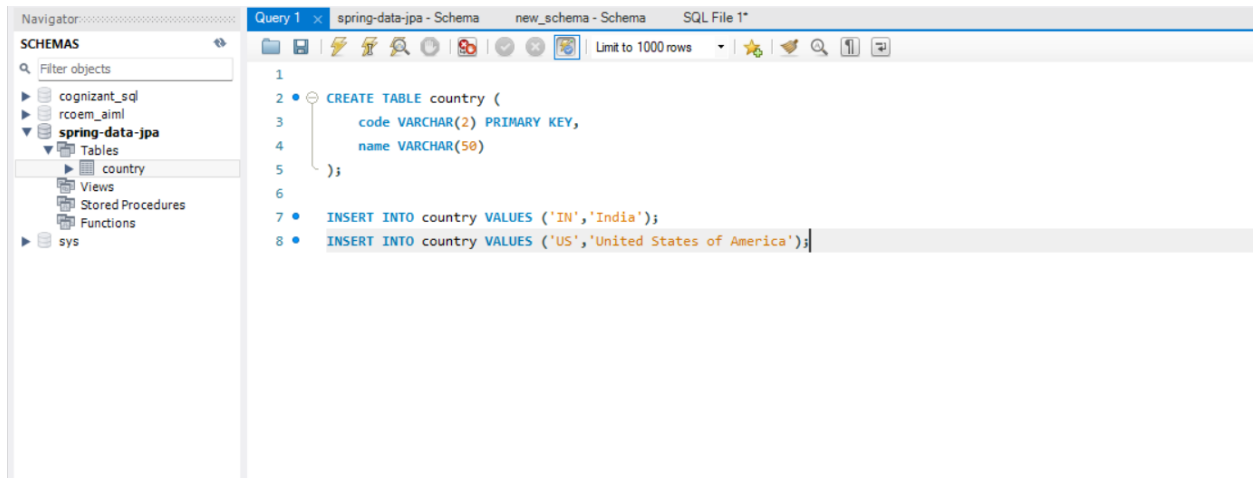
Create a Eclipse Project using Spring Initializr

- Go to <https://start.spring.io/>
- Change Group as "com.cognizant"
- Change Artifact Id as "orm-learn"
- In Options > Description enter "Demo project for Spring Data JPA and Hibernate"
- Click on menu and select "Spring Boot DevTools", "Spring Data JPA" and "MySQL Driver"
- Click Generate and download the project as zip
- Extract the zip in root folder to Eclipse Workspace
- Import the project in Eclipse "File > Import > Maven > Existing Maven Projects > Click Browse and select extracted folder > Finish"
- Create a new schema "ormlearn" in MySQL database. Execute the following commands to open MySQL client and create schema.

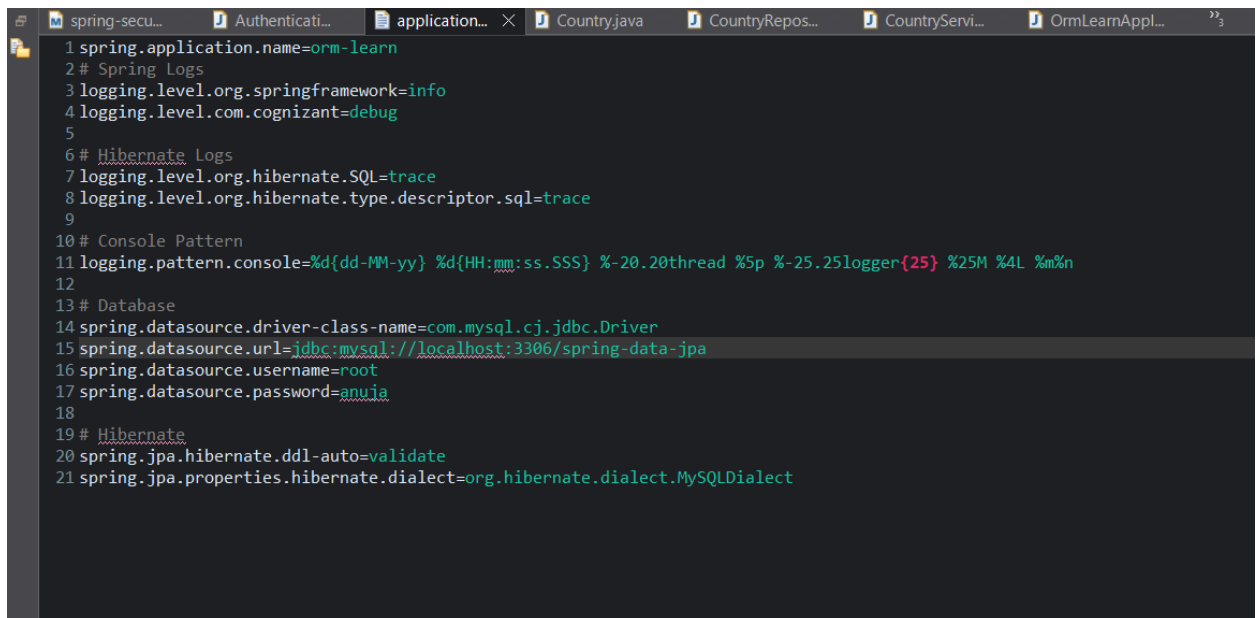


```
> mysql -u root -p
```

```
mysql> create schema ormlearn;
```



- In orm-learn Eclipse project, open `src/main/resources/application.properties` and include the below database and log configuration.



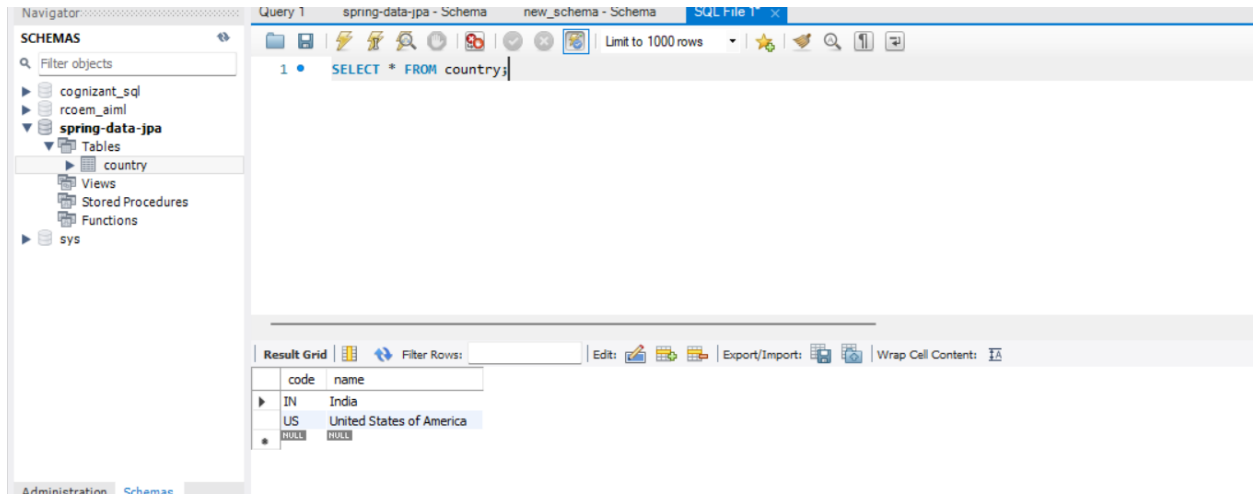
Country table creation

- Create a new table country with columns for code and name. For sample, let us insert one country with values 'IN' and 'India' in this table.

```
create table country(co_code varchar(2) primary key, co_name varchar(50));
```

- Insert couple of records into the table

```
insert into country values ('IN', 'India');  
insert into country values ('US', 'United States of America');
```



Persistence Class - com.cognizant.orm-learn.model.Country

```
1 package com.cognizant.orm_learn.model;
2 import jakarta.persistence.Column;
3 import jakarta.persistence.Entity;
4 import jakarta.persistence.Id;
5 import jakarta.persistence.Table;
6
7 @Entity
8 @Table(name = "country")
9 public class Country {
10
11     @Id
12     @Column(name = "code")
13     private String code;
14
15     @Column(name = "name")
16     private String name;
17
18     public Country() {
19     }
20
21     public Country(String code, String name) {
22         this.code = code;
23         this.name = name;
24     }
25
26     public String getCode() {
27         return code;
28     }
29
30     public void setCode(String code) {
31         this.code = code;
32     }
33
34     public String getName() {
35         return name;
36     }
37
38     public void setName(String name) {
39         this.name = name;
40     }
41
42     @Override
```

Repository Class - com.cognizant.orm-learn.CountryRepository

```
1 package com.cognizant.orm_learn.repository;
2
3
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import org.springframework.stereotype.Repository;
6
7 import com.cognizant.orm_learn.model.Country;
8
9 @Repository
10 public interface CountryRepository extends JpaRepository<Country, String> {
11
12 }
13
```

Service Class - com.cognizant.orm-learn.service.CountryService

```

1 package com.cognizant.orm_learn.service;
2
3 import java.util.List;
4
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.stereotype.Service;
7
8 import com.cognizant.orm_learn.model.Country;
9 import com.cognizant.orm_learn.repository.CountryRepository;
10
11 import jakarta.transaction.Transactional;
12
13 @Service
14 public class CountryService {
15
16     @Autowired
17     private CountryRepository countryRepository;
18
19     @Transactional
20     public List<Country> getAllCountries() {
21
22         return countryRepository.findAll();
23     }
24 }
25
26 }

```

- Execute main method to check if data from ormlearn database is retrieved.

```
File Edit View Window Help
package com.cognizant;

import java.util.List;

import org.springframework.boot.SpringApplication;
import org.springframework.context.annotation.Bean;

import com.cognizant.o.Repository;
import com.cognizant.o.TransactionRepository;
import jakarta.transaction.Transactional;

@Service
public class CountryService {

    @Autowired
    private CountryRepository countryRepo;

    @Transactional
    public List<Country> returnCountry() {
        // ... 
    }
}
```

Outline Task List Console

terminal - OrmLearnApplication [Java Application] C:\Users\anga\AppData\Local\Programs\Eclipse IDE\jdk-17.0.9-hotspot\bin\java.exe (14-Jul-2026, 8:49:32 pm ~ 8:49:36 pm elapsed 0:00:04) [pid: 23004]

:: Spring Boot :: (v4.1.0)

```
14-07-26 20:49:33.360 restartedMain INFO c.c.o.OrmLearnApplication logStarting 54 Starting OrmLearnApplication using Java 17.0.9 with PID 23004 (C:\
14-07-26 20:49:33.362 restartedMain DEBUG c.c.o.OrmLearnApplication logStarting 55 Running with Spring Boot v4.1.0, Spring v7.0.8
14-07-26 20:49:33.363 restartedMain INFO c.c.o.OrmLearnApplication logStartupProfileInfo 652 No active profile set, falling back to default profile: "default"
14-07-26 20:49:33.411 restartedMain INFO ertyDefaultsPostProcessor log 69 Devtools property defaults active! Set 'spring.devtools.add-proper
14-07-26 20:49:33.726 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 145 Bootstrapping Spring Data JPA repositories in DEFAULT mode.
14-07-26 20:49:33.787 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 213 Finished Spring Data repository scanning in 43 ms. Found 1 JPA rep
14-07-26 20:49:34.069 restartedMain INFO org.hibernate.orm.jpa logPersistenceUnitInformation 102 HHH0008540: Processing PersistenceUnitInfo [name: default]
14-07-26 20:49:34.121 restartedMain INFO org.hibernate.orm.core logVersion 41 HH0000001: Hibernate ORM core version 7.4.1.Final
14-07-26 20:49:34.368 restartedMain INFO SpringPersistenceUnitInfo addTransformer 129 No LoadTimeWeaver setup: ignoring JPA class transformer
14-07-26 20:49:34.706 restartedMain INFO c.z.h.HikariDataSource getConnection 109 HikariPool-1 - Starting...
14-07-26 20:49:35.015 restartedMain INFO c.z.h.pool.HikariPool checkIfTest 580 HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@7
14-07-26 20:49:35.015 restartedMain INFO c.z.h.HikariDataSource getConnection 122 HikariPool-1 - Start completed.
14-07-26 20:49:35.110 restartedMain WARN o.h.o.deprecation constructDialect 152 HH0000002: MySQLDialect does not need to be specified explicitly
14-07-26 20:49:35.124 restartedMain INFO o.h.o.connections.pooling logConnectionInfo 179 HH01001005: Database info:
Database JDBC URL [jdbc:mysql://localhost:3306/spring-data-jpa]
Database driver: MySQL Connector/J
Database dialect: MySQLDialect
Database version: 8.0.44
Default catalog/schema: spring-data-jpa/undefined
Autocommit mode: undefined/unknown
Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
JDBC fetch size: none
Pool: DataSourceConnectionFactory
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
14-07-26 20:49:35.887 restartedMain INFO org.hibernate.orm.core initiateService 53 HH0000489: No JTA platform available (set 'hibernate.transaction.j
14-07-26 20:49:35.872 restartedMain INFO EntityManagerFactoryBean buildInitiativeManagerFactory 464 Initialized JPA EntityManagerFactory for persistence unit 'd
14-07-26 20:49:35.956 restartedMain INFO q.QueryEnhancerFactories <init> 48 Hibernate is in classpath; If applicable, HQL parser will be used.
14-07-26 20:49:36.220 restartedMain INFO c.c.o.OrmLearnApplication logStarted 60 Started OrmLearnApplication in 3.218 seconds (process running fo
14-07-26 20:49:36.227 restartedMain INFO c.c.o.OrmLearnApplication main 27 Inside main
14-07-26 20:49:36.228 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 36 Start
14-07-26 20:49:36.405 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1_0.code,c1_0.name from country c1_0
14-07-26 20:49:36.442 restartedMain DEBUG c.c.o.OrmLearnApplication testGetAllCountries 40 Countries = [Country [code=IN, name=India], Country [code=US, name
14-07-26 20:49:36.443 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 42 End
14-07-26 20:49:36.451 locationShutdownHook INFO EntityManagerFactoryBean destroy 63 Closing JPA EntityManagerFactory for persistence unit 'default'
14-07-26 20:49:36.453 locationShutdownHook INFO c.z.h.HikariDataSource close 349 HikariPool-1 - Shutdown initiated...
14-07-26 20:49:36.451 locationShutdownHook INFO c.z.h.HikariDataSource close 353 HikariPool-1 - Shutdown completed.
```

Difference between JPA, Hibernate and Spring Data JPA

Java Persistence API (JPA)

- JSR 338 Specification for persisting, reading and managing data from Java objects
- Does not contain concrete implementation of the specification
- Hibernate is one of the implementation of JPA

Hibernate

- ORM Tool that implements JPA

Spring Data JPA

- Does not have JPA implementation, but reduces boiler plate code
- This is another level of abstraction over JPA implementation provider like Hibernate
- Manages transactions

Refer code snippets below on how the code compares between Hibernate and Spring Data JPA Hibernate

```
/* Method to CREATE an employee in the database */
public Integer addEmployee(Employee employee){
    Session session = factory.openSession();
    Transaction tx = null;
    Integer employeeID = null;

    try {
        tx = session.beginTransaction();
        employeeID = (Integer) session.save(employee);
        tx.commit();
    } catch (HibernateException e) {
        if (tx != null) tx.rollback();
        e.printStackTrace();
    }
}
```

```
    } finally {  
        session.close();  
    }  
    return employeeID;  
}
```

Spring Data JPA

EmployeeRepository.java

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
  
}
```

EmployeeService.java

```
@Autowired  
private EmployeeRepository employeeRepository;  
  
@Transactional  
public void addEmployee(Employee employee) {  
    employeeRepository.save(employee);  
}
```

Reference Links:

<https://dzone.com/articles/what-is-the-difference-between-hibernate-and-spring-1>

<https://www.javaworld.com/article/3379043/what-is-jpa-introduction-to-the-java-persistence-api.html>

